Lectures Review: Generative Learning Algorithms

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Outline

1. Generative Modeling
2. Gaussian Discriminant Analysis
3. Maximum Likelihood Fitting
4. Naive Bayes
5. Summary

Two ways to model the same data

Given i.i.d. pairs $(\mathbf{x}_n, y_n) \sim \mathcal{D}$ with $\mathbf{x}_n \in \mathbb{R}^d$ and labels $y_n \in \{1, \ldots, C\}$:

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The Central Question:

A generative model specifies forward sampling $(\mathbf{x}, y) \sim p(\mathbf{x}, y)$. Classification observes $\mathbf{x}$ and infers y.

Which mathematical operation reverses this direction, and what does it require?

From a joint model to an optimal decision

Posterior Inversion: For a fitted joint model $p(\mathbf{x}, y = c; \boldsymbol{\theta})$ with priors π_c and class densities $p(\mathbf{x} \mid y = c)$, Bayes' rule defines:

$$p(y = c \mid \mathbf{x}; \boldsymbol{\theta}) := \frac{p(\mathbf{x}, y = c; \boldsymbol{\theta})}{p(\mathbf{x}; \boldsymbol{\theta})}, \qquad p(\mathbf{x}; \boldsymbol{\theta}) = \sum_{k=1}^{C} \pi_k \, p(\mathbf{x} \mid y = k).$$

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Bayes Optimal Decision (0/1 Loss):

- Mislabling cost is 1; the error probability is $1 - p(y = \hat{c} \mid \mathbf{x};\, \boldsymbol{\theta})$.
- Minimum error risk selects the posterior mode: $\hat{y}^*(\mathbf{x}) = \arg\max_c p(y = c \mid \mathbf{x};\, \boldsymbol{\theta})$.

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Bayes' inversion turns generative models into computable decision scores,
with normalization guaranteeing calibrated probabilities.

Lemma: Posterior via weight normalization

Lemma (Bayes' Rule as Weight Normalization). The joint $w_c(\mathbf{x}) := \pi_c p(\mathbf{x} \mid y = c)$ acts as an unnormalized *class weight*. Dividing by total weight yields the posterior:

$$\Pr\left(y = c \mid \mathbf{x}\right) = \frac{\Pr\left(\mathbf{x}, y = c\right)}{\Pr(\mathbf{x})} = \frac{\pi_c \, p(\mathbf{x} \mid y = c)}{\sum_{k=1}^{C} \pi_k \, p(\mathbf{x} \mid y = k)}.$$

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Core Mechanism: The evidence $p(\mathbf{x}) = \sum_k w_k(\mathbf{x})$ acts as a shared partition function. Dividing by $p(\mathbf{x})$ guarantees $\sum_c p(y = c \mid \mathbf{x}) = 1$ while strictly preserving relative orderings.

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One shared total rescales all classes simultaneously:
posterior weights sum to 1, and their relative ordering is strictly preserved.

Decision Rule: Prediction drops the denominator

Because $\arg\max_c$ compares classes at fixed $\mathbf{x}$, the shared denominator cancels:

$$\hat{y}(\mathbf{x}) \in \arg\max_c \; p(y = c \mid \mathbf{x}; \boldsymbol{\theta}) \qquad \text{optimal rule under } 0/1 \text{ loss}$$

$$= \arg\max_c \; \frac{\pi_c \, p(\mathbf{x} \mid y = c)}{p(\mathbf{x}; \boldsymbol{\theta})} \qquad \text{Bayes' rule, previous lemma}$$

$$= \arg\max_c \; \pi_c \, p(\mathbf{x} \mid y = c) \qquad p(\mathbf{x}; \boldsymbol{\theta}) \text{ is class-independent: dropped}$$

$$= \arg\max_c \; \Big[\underbrace{\log \pi_c}_{\text{class prior}} + \underbrace{\log p(\mathbf{x} \mid y = c)}_{\text{feature match}} \Big] \qquad \log(\cdot) \text{ is strictly monotonically increasing}$$

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Dual Evaluation Modes:

- **For Classification:** Compare unnormalized scores $s_c(\mathbf{x}) := \log \pi_c + \log p(\mathbf{x} \mid y = c)$.
- **For Probabilities:** Evaluate softmax $p(y = c \mid \mathbf{x}) = \frac{e^{s_c(\mathbf{x})}}{\sum_k e^{s_k(\mathbf{x})}}$.

Decomposition: Denominator vs Prior

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To pick a label, compare unnormalized class weights (drop $p(\mathbf{x})$);
to state a probability, normalize them via softmax (keep $p(\mathbf{x})$).

Two likelihoods, one dataset

Both paradigms train on identical i.i.d. pairs $\mathcal{D} = \{(\mathbf{x}_n, y_n)\}_{n=1}^{N}$, but reward different objectives:

Discriminative: $L_{\text{cond}}(\mathbf{w}) = \prod_{n=1}^{N} p(y_n \mid \mathbf{x}_n;\, \mathbf{w})$ conditional label probability

Generative: $L_{\text{joint}}(\boldsymbol{\theta}) = \prod_{n=1}^{N} \pi_{y_n}\, p(\mathbf{x}_n \mid y_n;\, \boldsymbol{\theta})$ joint pair probability

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- **Capacity Allocation:** Conditional MLE optimizes solely for labels given $\mathbf{x}$; joint MLE must also explain input locations $p(\mathbf{x})$.
- **Test-Time Inference:** Both output $p(y = c \mid \mathbf{x})$; discriminative models evaluate a direct mapping, whereas generative models invert via Bayes' rule.

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A discriminative model assumes a boundary shape;
a generative model assumes a data-generating distribution.

Recap: The Generative Classification Pipeline

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4. **Next Step for Vector Inputs:** How do we specify continuous density $p(\mathbf{x} \mid y = c)$ on $\mathbb{R}^d$? Section 2 introduces Gaussian Discriminant Analysis (GDA).

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The Central Question:

Gaussian class models determine the functional form of the posterior.

Which boundary geometry do they dictate, and what assumption turns it linear?

The multivariate Gaussian density

For centroid $\boldsymbol{\mu} \in \mathbb{R}^d$ and symmetric positive definite $\boldsymbol{\Sigma} \in \mathbb{R}^{d \times d}$ ($\boldsymbol{\Sigma} \succ 0$):

Multivariate Gaussian Density.

$$\mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{d/2} |\boldsymbol{\Sigma}|^{1/2}} \exp\left[-\tfrac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right].$$

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Geometric Decomposition:

- **Elliptical Contours:** Mean $\boldsymbol{\mu}$ sets the center of the level sets.
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- **Regularity Requirement:** $\boldsymbol{\Sigma} \succ 0$ (strictly positive definite) guarantees full volume $|\boldsymbol{\Sigma}| > 0$; singular $\boldsymbol{\Sigma}$ collapses density to a lower-dimensional subspace.

GDA: the model and its class scores

Gaussian Discriminant Analysis (GDA). For C classes with categorical prior $\pi_c > 0$, $\sum_c \pi_c = 1$:

$$y \sim \text{Categorical}(\pi_1, \ldots, \pi_C), \qquad \mathbf{x} \mid y = c \sim \mathcal{N}(\boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c).$$

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Log-Class Score: Substitute the Gaussian density into class score $s_c(\mathbf{x}) = \log[\pi_c p(\mathbf{x} \mid y = c)]$:

$$s_c(\mathbf{x}) = \underbrace{\log \pi_c}_{\text{class prior}} - \underbrace{\tfrac{1}{2} \log|\boldsymbol{\Sigma}_c|}_{\text{volume penalty}} - \underbrace{\tfrac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_c)^\top \boldsymbol{\Sigma}_c^{-1}(\mathbf{x} - \boldsymbol{\mu}_c)}_{\text{squared Mahalanobis distance}} - \tfrac{d}{2} \log(2\pi)$$

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Decision Surface: The pairwise boundary between classes c and k is the level set $s_c(\mathbf{x}) = s_k(\mathbf{x})$: a difference of two quadratic forms in $\mathbf{x}$.

QDA: class-specific covariance gives curved boundaries

With class-specific $\boldsymbol{\Sigma}_c$, expand the boundary condition $s_c(\mathbf{x}) - s_k(\mathbf{x}) = 0$:

$$\underbrace{\mathbf{x}^\top \left(\boldsymbol{\Sigma}_c^{-1} - \boldsymbol{\Sigma}_k^{-1}\right)\mathbf{x}}_{\text{Quadratic Term: Curvature}} \quad -2\,\mathbf{x}^\top \left(\boldsymbol{\Sigma}_c^{-1}\boldsymbol{\mu}_c - \boldsymbol{\Sigma}_k^{-1}\boldsymbol{\mu}_k\right) + \underbrace{\log \frac{|\boldsymbol{\Sigma}_c|}{|\boldsymbol{\Sigma}_k|} + \cdots}_{\text{constants}} = 0.$$

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- Because $\boldsymbol{\Sigma}_c^{-1} - \boldsymbol{\Sigma}_k^{-1} \neq \mathbf{0}$, the quadratic term survives.
- Pairwise boundaries are quadric hypersurfaces (hyperbolas, ellipses, parabolas in $\mathbb{R}^2$).
- Highly expressive, but requires estimating $C \cdot \frac{d(d+1)}{2}$ covariance elements.

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Curved boundaries are the mathematical signature of unequal class covariances.

LDA: one shared covariance kills the quadratic term

Homoscedasticity ($\boldsymbol{\Sigma}_c = \boldsymbol{\Sigma}$): Expand the quadratic form to track class dependency:

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Class-free terms $-\frac{1}{2} \log|\boldsymbol{\Sigma}|$ and $-\frac{d}{2} \log(2\pi)$ also cancel identically in differences $s_c(\mathbf{x}) - s_k(\mathbf{x})$.

LDA: one shared covariance kills the quadratic term

Homoscedasticity ($\boldsymbol{\Sigma}_c = \boldsymbol{\Sigma}$): Expand the quadratic form to track class dependency:

$$
\begin{aligned}
(\mathbf{x} - \boldsymbol{\mu}_c)^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}_c) = \quad & \mathbf{x}^\top \boldsymbol{\Sigma}^{-1} \mathbf{x} && \text{Class-free: identical for every class } c \\
& - 2\, \mathbf{x}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_c && \textbf{Linear in } \mathbf{x}\text{: carries class index } c \\
& + \boldsymbol{\mu}_c^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_c && \textbf{Constant in } \mathbf{x}\text{: carries class index } c
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Class-free terms $-\frac{1}{2} \log|\boldsymbol{\Sigma}|$ and $-\frac{d}{2} \log(2\pi)$ also cancel identically in differences $s_c(\mathbf{x}) - s_k(\mathbf{x})$.

With shared covariance, all quadratic dependence on $\mathbf{x}$ cancels out identically.
Decision boundaries become linear hyperplanes: **Linear Discriminant Analysis (LDA)**.

Conclusion: the linear discriminant function

Discarding the class-free remainder $\kappa(\mathbf{x}) := -\frac{1}{2}\mathbf{x}^\top \boldsymbol{\Sigma}^{-1}\mathbf{x} - \frac{1}{2}\log|\boldsymbol{\Sigma}| - \frac{d}{2}\log(2\pi)$:

$$\boldsymbol{\beta}_c := \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_c, \qquad \gamma_c := \log \pi_c - \tfrac{1}{2}\boldsymbol{\mu}_c^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_c \qquad \text{canonical weights and bias}$$

$$\delta_c(\mathbf{x}) := \mathbf{x}^\top \boldsymbol{\beta}_c + \gamma_c, \qquad \hat{y}(\mathbf{x}) \in \arg\max_c \delta_c(\mathbf{x}) \qquad \text{linear discriminant score}$$

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Softmax Posterior: Canceling shared $\kappa(\mathbf{x})$ yields a softmax with linear scores:

$$p(y = c \mid \mathbf{x}) = \frac{\exp(\mathbf{x}^\top\boldsymbol{\beta}_c + \gamma_c)}{\sum_{k=1}^{C}\exp(\mathbf{x}^\top\boldsymbol{\beta}_k + \gamma_k)}.$$

Binary LDA recovers the sigmoid

For binary classification ($C = 2$, labels $\{0, 1\}$), the log-posterior odds expand to:

$$\eta(\mathbf{x}) := \log \frac{p(1 \mid \mathbf{x})}{p(0 \mid \mathbf{x})} = \delta_1(\mathbf{x}) - \delta_0(\mathbf{x}) = \mathbf{x}^\top \mathbf{w} + b$$

$$\mathbf{w} = \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)$$ weight vector: normal to hyperplane

$$b = -\tfrac{1}{2}\left(\boldsymbol{\mu}_1^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_1 - \boldsymbol{\mu}_0^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_0\right) + \log \tfrac{\pi_1}{\pi_0}$$ bias scalar: boundary spatial offset

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Hence $p(1 \mid \mathbf{x}) = \sigma(\mathbf{x}^\top \mathbf{w} + b)$ with logistic sigmoid $\sigma(\eta) = \frac{1}{1+e^{-\eta}}$. The posterior matches logistic regression, with parameters derived from Gaussian moments.

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Linear log-odds are a mathematical theorem of shared-covariance Gaussian classes, not an ad-hoc extra assumption.

Geometry: where the boundary sits

Predict $y = 1 \iff \eta(\mathbf{x}) = \mathbf{x}^\top \mathbf{w} + b > 0$. Three geometric cases illuminate the boundary's behavior:

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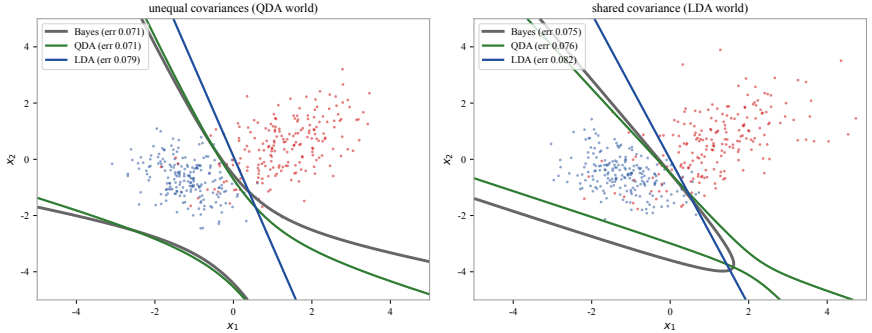
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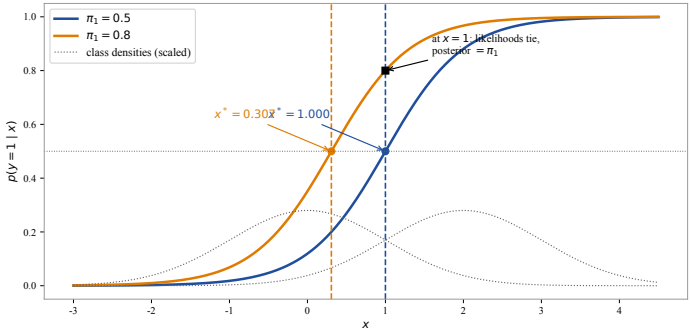
Class means and covariance set the boundary orientation ($\mathbf{w}$);
class priors set its spatial translation offset (b).

Living case: measured QDA and LDA boundaries



With unequal true covariances, fitted QDA tracks the curved Bayes boundary while LDA pays for its linear constraint; with truly shared covariance the two coincide.

Living case: the prior moves the boundary, measured



With $\mu_0 = 0$, $\mu_1 = 2$, shared variance 1, raising π_1 from 0.5 to 0.8 moves the measured threshold from $x = 1$ to $x \approx 0.31$; at $x = 1$ the likelihoods tie and the posterior equals the prior.

Recap: from Gaussian classes to boundaries

1. **Model Structure:** $y \sim \text{Categorical}(\pi)$ and $\mathbf{x} \mid y{=}c \sim \mathcal{N}(\boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c)$. The log-score $s_c(\mathbf{x}) = \log \pi_c + \log \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c)$ is quadratic in $\mathbf{x}$.

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4. **Geometric Partitioning:** Centroids and covariance control hyperplane normal $\mathbf{w}$; prior ratio π_1/π_0 controls spatial offset b.

Outline

1. Generative Modeling
2. Gaussian Discriminant Analysis
3. Maximum Likelihood Fitting
4. Naive Bayes
5. Summary

The need: estimate θ from labeled pairs

Problem Setup: Generative parameters $\boldsymbol{\theta} = \big(\{\pi_c\}, \{\boldsymbol{\mu}_c\}, \{\boldsymbol{\Sigma}_c\}\big)$ are unknown and must be learned from training pairs $\mathcal{D} = \{(\mathbf{x}_n, y_n)\}_{n=1}^{N}$.

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Joint Maximum Likelihood Estimation (MLE): Maximize the joint probability of the observed pairs:

$$\hat{\boldsymbol{\theta}}_{\text{MLE}} = \arg\max_{\boldsymbol{\theta}} \sum_{n=1}^{N} \log p(\mathbf{x}_n, y_n; \boldsymbol{\theta}).$$

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The Central Question:

Does the joint MLE admit an exact closed form?

Why can LDA and logistic regression, sharing the same posterior family, fit *different* boundaries on identical data?

The formal question: a likelihood that splits by class

Let $N_c := \sum_{n=1}^{N} \mathbf{1}\{y_n = c\}$ count training points in class c. Factorizing the joint likelihood $p(\mathbf{x}, y = c) = \pi_c \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c)$:

$$L(\boldsymbol{\theta}) = \prod_{n=1}^{N} \pi_{y_n} \mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_{y_n}, \boldsymbol{\Sigma}_{y_n}) \qquad \text{i.i.d. observations}$$

$$\log L(\boldsymbol{\theta}) = \underbrace{\sum_{c=1}^{C} N_c \log \pi_c}_{\text{Prior Term: depends only on counts } N_c} + \underbrace{\sum_{c=1}^{C} \sum_{n:\, y_n = c} \log \mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c)}_{\text{Conditional Density Term: decoupled by class}}$$

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Decoupling Property of Fully Observed Labels: Observed labels partition the dataset into independent subsets; each class's Gaussian parameters appear strictly in its own sum.

Lemma: the prior MLE is the class frequency

Optimization Problem: Maximize $\sum_{c=1}^{C} N_c \log \pi_c$ subject to $\sum_{c=1}^{C} \pi_c = 1$.

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$$\frac{\partial \mathcal{L}}{\partial \pi_c} = \frac{N_c}{\pi_c} - \lambda = 0 \quad \Longrightarrow \quad \pi_c = \frac{N_c}{\lambda} \quad \xrightarrow{\sum \pi_c = 1} \quad \hat{\pi}_c = \frac{N_c}{N}$$

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Class priors depend strictly on label counts N_c;
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Let $\mathbf{r}_n := \mathbf{x}_n - \boldsymbol{\mu}_c$. Differentiating the class-c log-likelihood with respect to centroid $\boldsymbol{\mu}_c$:

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Moment Matching Principle: Maximum likelihood matches the theoretical distribution mean to the empirical sample mean, extending the 1D sample mean to vector centroids in $\mathbb{R}^d$.

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Fix class c, let precision $\mathbf{A} := \boldsymbol{\Sigma}^{-1} \succ 0$ and residuals $\hat{\mathbf{r}}_n := \mathbf{x}_n - \hat{\boldsymbol{\mu}}_c$. Objective:
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Conclusion: The covariance MLE is the average outer product of residuals: the matrix generalization of sample variance.

Conclusion: QDA and LDA covariance estimates

QDA (Class-Specific) vs LDA (Shared Pooled Scatter): Using class-centered residuals $\hat{\mathbf{r}}_n := \mathbf{x}_n - \hat{\boldsymbol{\mu}}_{y_n}$:

$$\textbf{QDA:} \quad \hat{\boldsymbol{\Sigma}}_c = \frac{1}{N_c} \sum_{n:\, y_n = c} \hat{\mathbf{r}}_n \hat{\mathbf{r}}_n^\top \qquad \text{vs} \qquad \textbf{LDA:} \quad \hat{\boldsymbol{\Sigma}} = \frac{1}{N} \sum_{n=1}^{N} \hat{\mathbf{r}}_n \hat{\mathbf{r}}_n^\top .$$

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Structural Distinctions:

- **Centering Rule:** Both subtract *each point's own class mean* $\hat{\boldsymbol{\mu}}_{y_n}$, **never** the global sample mean.
- **Pooled Average:** LDA pools all N residuals, weighting classes proportionally: $\hat{\boldsymbol{\Sigma}} = \sum_c \frac{N_c}{N} \hat{\boldsymbol{\Sigma}}_c$.
- **Parameter Economy:** QDA fits $C \cdot \frac{d(d+1)}{2}$ parameters; LDA fits only $\frac{d(d+1)}{2}$.

Counterfactual: Singular scatter and regularization

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- **High-Dimensional Regime** ($N \leq d$)**:** Residuals span at most $N - C$ dimensions $\Longrightarrow |\hat{\mathbf{\Sigma}}| = 0$, log-likelihood diverges to $-\infty$.
- **Collinear Features:** Redundant features make $\hat{\mathbf{\Sigma}}$ singular even when $N \mathbf{g} d$.

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- **High-Dimensional Regime** ($N \leq d$)**:** Residuals span at most $N - C$ dimensions $\Longrightarrow |\hat{\mathbf{\Sigma}}| = 0$, log-likelihood diverges to $-\infty$.
- **Collinear Features:** Redundant features make $\hat{\mathbf{\Sigma}}$ singular even when $N \mathbf{g} d$.
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Solve the linear system $\hat{\boldsymbol{\Sigma}}_\lambda \mathbf{w} = \hat{\boldsymbol{\mu}}_1 - \hat{\boldsymbol{\mu}}_0$ via Cholesky decomposition;
never form the explicit inverse $\hat{\boldsymbol{\Sigma}}^{-1}$.

The Likelihood Split: Why Objectives Disagree

Both models share posterior $\sigma(\mathbf{x}^\top \mathbf{w} + b)$, but optimize different objectives:

$$\log L_{\text{joint}}(\boldsymbol{\theta}) = \underbrace{\sum_{n=1}^{N} \log p(y_n \mid \mathbf{x}_n;\, \boldsymbol{\theta})}_{\text{Conditional: Classification}} + \underbrace{\sum_{n=1}^{N} \log p(\mathbf{x}_n;\, \boldsymbol{\theta})}_{\text{Marginal: Density Match}} .$$

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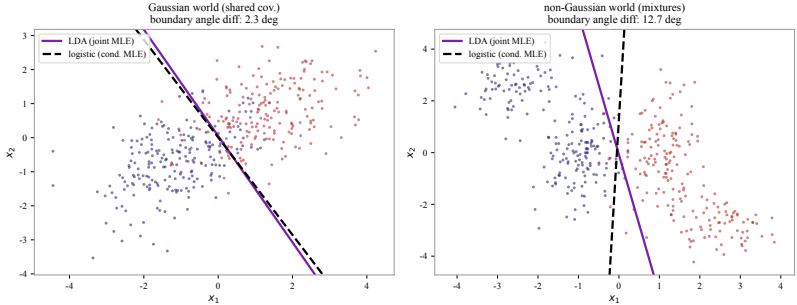
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The posterior family specifies how the model predicts;
the likelihood objective specifies how its parameters are learned.

Living case: same family, different boundaries, measured



In a Gaussian world the LDA and logistic boundaries nearly coincide; in a non-Gaussian world, joint and conditional fitting select visibly different $(\mathbf{w}, b)$.

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4. **Likelihood Decomposition:** $\log L_{\text{joint}} = \log L_{\text{cond}} + \sum_n \log p(\mathbf{x}_n)$. The marginal input term explains why joint and conditional estimators diverge off the Gaussian manifold.

Outline

1. Generative Modeling
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3. Maximum Likelihood Fitting
4. Naive Bayes
5. Summary

The need: text breaks the Gaussian

In text classification, a document is encoded as a binary presence vector $\mathbf{x} \in \{0,1\}^d$ over a dictionary of d words: $x_j = \mathbf{1}\{\text{word } j \text{ appears}\}$.

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The Combinatorial Curse of Dimensionality:

- The feature space $\{0,1\}^d$ contains 2^d distinct binary configurations.
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The Central Question:

What structural assumption reduces $2^d - 1$ parameters to $O(d)$,
and what does that assumption claim about how words appear?

The formal question: one Bernoulli per word

Per-Feature Model: For feature index j and class c, let $q_{jc} := p(x_j = 1 \mid y = c) \in [0, 1]$ denote word occurrence probability:

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Exact Chain Rule Decomposition (No Assumptions Yet):

$$p(\mathbf{x} \mid y = c) = \prod_{j=1}^{d} p\big(x_j \mid x_1, \ldots, x_{j-1}, y = c\big).$$

Each factor conditions on all previous words, preserving all $2^d - 1$ parameters.

The assumption: conditional independence given the class

Naive Bayes Conditional Independence Assumption. Features are mutually independent *conditional on the class label*:

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The model is naive about word dependencies, not about labels:
$2d$ linear parameters replace $2^d - 1$ combinatorial parameters.

Lemma: the MLE of each word probability is a fraction

Decoupled Joint Likelihood:

With $N_c := \sum_n \mathbf{1}\{y_n = c\}$ and $N_{jc} := \sum_{n: y_n = c} x_{nj}$ counting word j in class c:

$$L(\boldsymbol{\pi}, \mathbf{Q}) = \prod_{c=1}^{C} \pi_c^{N_c} \prod_{j=1}^{d} q_{jc}^{N_{jc}} (1 - q_{jc})^{N_c - N_{jc}}.$$

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Stationarity for Word Parameter q_{jc}:

$$\frac{\partial \log L}{\partial q_{jc}} = \frac{N_{jc}}{q_{jc}} - \frac{N_c - N_{jc}}{1 - q_{jc}} = 0 \quad \Longrightarrow \quad \hat{q}_{jc} = \frac{N_{jc}}{N_c}$$

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Counting MLE Principle: The MLE matches q_{jc} to the empirical sample proportion of word j in class c. Fitting requires only a single pass of counting over the training corpus.

Scoring a message: every word selects a factor

Additive Log-Class Score Formulation:

The log-score for class c is an additive sum over vocabulary terms:

$$s_c(\mathbf{x}) := \log \hat{\pi}_c + \sum_{j=1}^{d} \Big[x_j \log \hat{q}_{jc} + (1 - x_j) \log(1 - \hat{q}_{jc}) \Big].$$

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Key Structural Mechanics:

- **Word Presence** ($x_j = 1$)**:** Selects factor $\log \hat{q}_{jc}$.
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- **Numerical Stability:** Log-domain addition prevents floating-point underflow (e^{-d}).
- **Posterior Sigmoid:** In binary classification, $p(1 \mid \mathbf{x}) = \sigma(s_1(\mathbf{x}) - s_0(\mathbf{x}))$.

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The Zero-Count Problem: Suppose word j never appears in class c in training ($N_{jc} = 0 \implies \hat{q}_{jc} = 0$).

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A single unseen word can erase the accumulated evidence of thousands of words; robust estimation requires strictly positive probabilities for all outcomes.

Laplace smoothing: a pseudo-count per outcome

Laplace Smoothing (Categorical Likelihood). For K possible outcomes with counts $M_1, \ldots, M_K$ ($\sum_k M_k = M$):

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- Each feature x_j has $K = 2$ outcomes: presence ($x_j = 1$) and absence ($x_j = 0$).
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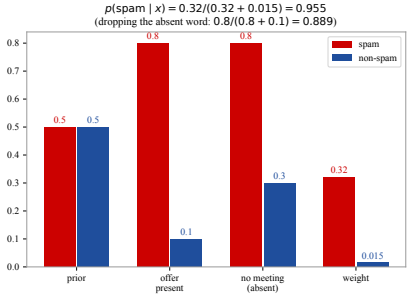
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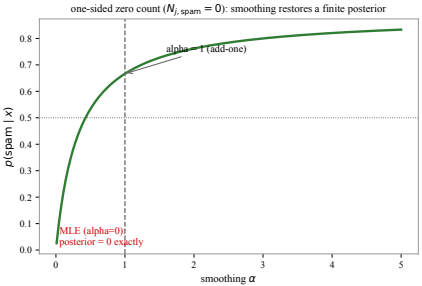
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- **Add-α Generalization:** $\tilde{q}_{jc} = \frac{N_{jc}+\alpha}{N_c+2\alpha}$ (symmetric Beta(α, α) prior).

Living case: a two-word message, computed and smoothed





Left: weights 0.32 vs 0.015 give $p(\text{spam}) \approx 0.955$, and dropping the absent-word factor gives 0.889.

Right: with a one-sided zero count the MLE posterior is exactly 0; any $\alpha > 0$ restores finite evidence.

Recap: independence, counting, and smoothing

1. **Class-Conditional Independence:** $p(\mathbf{x} \mid y = c) = \prod_{j=1}^{d} q_{jc}^{x_j}(1 - q_{jc})^{1-x_j}$. Collapses parameter complexity from $2^d - 1$ to $2d$.

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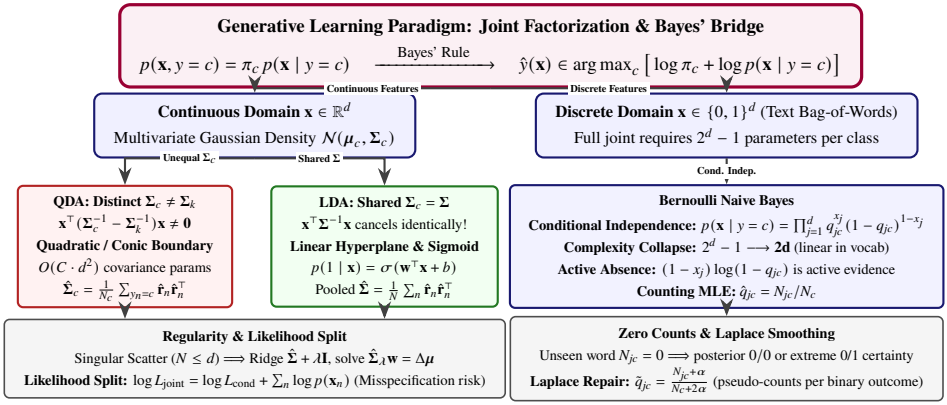
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4. **Laplace Regularization:** Zero counts produce numerical breakdown or extreme certainty; Laplace smoothing assigns pseudo-counts to preserve positive probability everywhere.

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Grand Epilogue: Landscape of Generative Learning



Q1 (1/2): What Does a Generative Model Buy? — The Bayes' Bridge

Both paradigms train on pairs $(\mathbf{x}_n, y_n)$; the difference is *which distribution the likelihood rewards*.

Generative pipeline and Bayes' bridge. Model the joint $p(\mathbf{x}, y{=}c) = \pi_c\, p(\mathbf{x} \mid y{=}c)$, then invert for classification:

$$p(y{=}c \mid \mathbf{x}) = \frac{\pi_c\, p(\mathbf{x} \mid y{=}c)}{\sum_k \pi_k\, p(\mathbf{x} \mid y{=}k)}, \qquad \hat{y}(\mathbf{x}) \in \arg\max_c \Big[\log \pi_c + \log p(\mathbf{x} \mid y{=}c) \Big].$$

The denominator $p(\mathbf{x})$ is identical across candidate classes and drops from the arg max; the prior π_c carries the class index and must survive.

To pick a label, compare the class weights (drop the denominator);
to state a calibrated probability, normalize them (keep the denominator).

Q1 (2/2): What Does a Generative Model Buy? — Asymmetry of Assumptions

Neither paradigm is assumption-free; each commits to a different target:

Discriminative vs Generative Assumptions.

- **Discriminative (linear log-odds):** assumes $\log \frac{p(1|\mathbf{x})}{p(0|\mathbf{x})} = \mathbf{x}^\top \mathbf{w} + b$. Imposes no shape on input class distributions.
- **Generative (Gaussian classes):** assumes $\mathbf{x} \mid y{=}c \sim \mathcal{N}(\boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c)$. The boundary is derived from the data shape, not chosen directly.

A discriminative model assumes a boundary shape; a generative model assumes a data-generating shape. The generative assumption is stronger: it excels under scarce data, while misspecification distorts the posterior everywhere. Under model mismatch, discriminative fitting is the safer choice.

Q2 (1/2): Why Does Shared Covariance Linearize? — QDA & Curvature

GDA scores each class with $s_c(\mathbf{x}) = \log \pi_c - \frac{1}{2}\log|\boldsymbol{\Sigma}_c| - \frac{1}{2}(\mathbf{x}-\boldsymbol{\mu}_c)^\top \boldsymbol{\Sigma}_c^{-1}(\mathbf{x}-\boldsymbol{\mu}_c) - \frac{d}{2}\log 2\pi$.

Class-specific covariance (QDA): curved boundaries. Setting $s_c(\mathbf{x}) = s_k(\mathbf{x})$ leaves the quadratic difference term alive:

$$\mathbf{x}^\top\left(\boldsymbol{\Sigma}_c^{-1} - \boldsymbol{\Sigma}_k^{-1}\right)\mathbf{x} - 2\,\mathbf{x}^\top\left(\boldsymbol{\Sigma}_c^{-1}\boldsymbol{\mu}_c - \boldsymbol{\Sigma}_k^{-1}\boldsymbol{\mu}_k\right) + (\text{constants}) = 0.$$

Pairwise decision surfaces are conic sections (hyperbolas, ellipses, parabolas).

Curved boundaries are the mathematical signature of unequal class covariances.
QDA tracks curved Bayes boundaries, but requires estimating $O(C \cdot d^2)$ parameters.

Q2 (2/2): Why Does Shared Covariance Linearize? — LDA & Cancellation

Assuming $\boldsymbol{\Sigma}_c = \boldsymbol{\Sigma}$ for all classes, the quadratic term $\mathbf{x}^\top \boldsymbol{\Sigma}^{-1} \mathbf{x}$, the log-determinant, and $\frac{d}{2} \log 2\pi$ become class-free and cancel out in every comparison.

LDA: linear discriminants and Sigmoid recovery.

$$\delta_c(\mathbf{x}) = \gamma_c + \mathbf{x}^\top \boldsymbol{\beta}_c, \quad \boldsymbol{\beta}_c = \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_c, \quad \gamma_c = \log \pi_c - \tfrac{1}{2} \boldsymbol{\mu}_c^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_c.$$

In the binary case: $p(1 \mid \mathbf{x}) = \sigma(\mathbf{x}^\top \mathbf{w} + b)$ with $\mathbf{w} = \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)$.

Linear log-odds are a theorem of shared covariance, not an extra assumption.
The means and $\boldsymbol{\Sigma}$ set the boundary's direction ($\mathbf{w}$); the prior ratio $\log(\pi_1/\pi_0)$ sets its offset b (measured: $x^* = 1 \to 0.31$).

Q3 (1/2): What Does Joint MLE Fit? — Closed-Form Estimates

Observed labels partition the joint likelihood by class, giving instant closed forms.

GDA / LDA maximum likelihood closed forms. With N_c points in class c and residuals $\hat{\mathbf{r}}_n = \mathbf{x}_n - \hat{\boldsymbol{\mu}}_{y_n}$:

$$\hat{\pi}_c = \frac{N_c}{N}, \qquad \hat{\boldsymbol{\mu}}_c = \frac{1}{N_c} \sum_{n:\, y_n = c} \mathbf{x}_n, \qquad \hat{\boldsymbol{\Sigma}} = \frac{1}{N} \sum_{n=1}^{N} \hat{\mathbf{r}}_n \hat{\mathbf{r}}_n^\top \text{ (LDA pooled).}$$

Residuals are centered at each point's *own class mean*, never the global mean. If scatter is singular ($N \le d$), ridge via $\hat{\boldsymbol{\Sigma}} + \lambda \mathbf{I}$ and solve $\hat{\boldsymbol{\Sigma}}_\lambda \mathbf{w} = \Delta\boldsymbol{\mu}$.

Each estimate is moment matching: categorical means to label counts, Gaussian means to class averages, and scatter to within-class residuals.

Q3 (2/2): What Does Joint MLE Fit? — The Likelihood Split

LDA and logistic regression share the same posterior family $\sigma(\mathbf{x}^\top \mathbf{w} + b)$, yet maximize different objectives:

Joint = conditional fitting + input-fitting term.

$$\log L(\boldsymbol{\theta}) = \underbrace{\sum_{n=1}^{N} \log p(y_n \mid \mathbf{x}_n;\, \boldsymbol{\theta})}_{\text{conditional fitting term (logistic)}} + \underbrace{\sum_{n=1}^{N} \log p(\mathbf{x}_n;\, \boldsymbol{\theta})}_{\text{input-fitting term (LDA)}} .$$

The same $\boldsymbol{\theta}$ controls both terms: joint MLE must explain where inputs sit, while conditional MLE focuses solely on separating labels.

The sigmoid says how the model predicts; the likelihood says how it learns.
Off the Gaussian manifold, the input term drags the boundary (measured angle diff: 12.7°).

Q4 (1/2): Why Does Naive Bayes Work? — Independence & Counting

Text features live in $\{0,1\}^d$; an unconstrained model requires $2^d - 1$ parameters per class.

Bernoulli Naive Bayes and counting MLE. Assume features are independent conditional on the class:

$$p(\mathbf{x} \mid y{=}c) = \prod_{j=1}^{d} q_{jc}^{x_j}(1 - q_{jc})^{1-x_j}, \qquad \hat{q}_{jc} = \frac{N_{jc}}{N_c}.$$

Independence is claimed within a class, so words may correlate through the label. An absent word contributes $\log(1 - \hat{q}_{jc})$, acting as an active evidence factor.

Naive Bayes collapses $2^d - 1$ parameters to $2d$ by assuming class-conditional independence.
Absence is evidence: dropping absent words shifts posterior confidence (measured: $0.955 \rightarrow 0.889$).

Q4 (2/2): Why Does Naive Bayes Work? — Zero Counts & Laplace Repair

If a word was never observed in training ($N_{jc} = 0$), encountering it in test data zeros out whole class weights, resulting in an undefined $0/0$ or extreme $0/1$ certainty.

Laplace smoothing (add-α pseudo-counts). A Bernoulli variable has two outcomes (presence and absence), so each receives a ghost count:

$$\tilde{q}_{jc} = \frac{N_{jc} + 1}{N_c + 2} \quad \text{or} \quad \tilde{q}_{jc} = \frac{N_{jc} + \alpha}{N_c + 2\alpha} \quad (\alpha > 0 \text{ tuned on validation}).$$

Every factor remains strictly positive, restoring finite posterior evidence for unseen words.

Naive Bayes is Bernoulli MLE scored in log space to prevent underflow,
and smoothed per outcome (+2 in denominator) so no single unvisited word can erase all evidence.